

JanGyan — Learn Without Limits

Category : Cyber Safety & Digital Security

Module 2 - Digital Identity & Privacy

Module Overview

This handbook is your complete, beginner-to-advanced guide to **Digital Identity & Privacy** — what personal data really means, how to control your privacy settings, how your data gets shared (often without you realising it), and what a "digital footprint" is and why it never truly disappears. Whether you are a student, a working professional, a parent, or a senior citizen, this handbook will help you understand and protect the most valuable thing you own online: your identity.

By the end of this module, you will not need to search anywhere else to understand the fundamentals of digital identity and privacy.

Learning Objectives

After completing this module, you will be able to:

- Define personal data, sensitive personal data, and digital identity
- Understand the core principles that govern responsible data handling
- Configure privacy settings across your phone, apps, and social media
- Recognise how and why your data gets shared with third parties
- Understand what a digital footprint is and how it can affect your future
- Know your rights under India's data protection law
- Apply practical privacy habits in daily digital life
- Know exactly where and how to exercise your data rights or file a complaint

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Why This Topic Matters

India had over **1.03 billion internet users** and around **500 million social media user identities** by the end of 2025. Every search, every app download, every "Accept All Cookies" click, and every photo posted online adds another layer to your digital identity — a layer that companies, advertisers, and sometimes criminals can access.

> *Reality Check: A 2025 survey found that while 60% of parents claimed awareness of online privacy risks like identity theft, only about 43% regularly adjusted privacy settings on their or their children's devices — and nearly one-third never touched their privacy settings at all.*

In 2017, India's Supreme Court delivered a landmark, unanimous nine-judge verdict declaring privacy a **fundamental right** under Article 21 of the Constitution. This means your privacy is not a favour companies or the government grant you — it is a right you already have. But rights only protect you when you know how to use them.

Introduction

Digital Identity is the sum of all the data points that represent you online — your Aadhaar and PAN details, your social media profiles, your email accounts, your browsing habits, your device IDs, your location history, and even your biometric data (fingerprints, face scans).

Privacy, in the digital context, means having control over who can see, collect, store, use, or share this identity — and being able to say no.

Think of your digital identity like a house built entirely of glass. Every action you take online adds another window. Privacy settings are the curtains — they don't stop the house from existing, but they let you decide who gets to look inside, and when.

Fundamental Concepts

Personal Data vs Sensitive Personal Data vs Digital Identity

Term	Meaning	Example
Personal Data	Any data that identifies you, directly or indirectly	Name, phone number, email, address
Sensitive Personal Data	Personal data requiring extra protection due to higher risk if misused	Biometric data, health records, financial details, passwords
Digital Identity	The complete online representation built from all your personal data over time	Your combined social media, banking, government ID, and browsing profile

Core Privacy Principles

Modern data protection laws worldwide — including India's DPDP Act, 2023 — are built on a common set of principles:

Principle	Simple Meaning
Notice	You must be clearly told what data is being collected and why
Consent	Your data can only be used if you have freely and clearly agreed
Purpose Limitation	Data can only be used for the reason it was originally collected
Data Minimisation	Only the data actually necessary should be collected
Storage Limitation	Data should not be kept longer than needed
Security Safeguards	Reasonable technical measures must protect your data from breach
Accountability	Organisations must be able to demonstrate compliance, not just claim it

Key Roles Under India's Data Protection Framework

- **Data Principal** — You, the individual to whom the personal data belongs

- **Data Fiduciary** — The organisation or person that decides why and how your data is processed (e.g., an app, a bank, a hospital)
- **Data Processor** — An entity that processes data on behalf of a Data Fiduciary
- **Consent Manager** — A registered, independent platform through which you can give, manage, review, or withdraw your consent
- **Data Protection Board of India** — The independent body that hears complaints and penalises violations

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Complete Detailed Explanation

Part A — Categories of Personal Data

Category	Examples
Identity Data	Name, Aadhaar number, PAN, passport, date of birth
Contact Data	Phone number, email address, home address
Financial Data	Bank account, UPI ID, credit/debit card details, income
Biometric Data	Fingerprints, iris scans, facial recognition data
Health Data	Medical records, prescriptions, fitness app data
Location Data	GPS location, travel history, check-ins
Behavioural Data	Search history, app usage, purchase history, likes/follows
Device Data	IP address, device ID, browser fingerprint, cookies

Part B — Privacy Settings, Platform by Platform

- **Smartphone Permissions (Android/iOS):** Every app requests permissions (camera, microphone, location, contacts, storage). Review these under Settings → Apps → Permissions, and revoke anything not essential to the app's core function.
- **Google Account Privacy Checkup:** Google's Privacy Checkup tool lets you review what data is saved (location history, web & app activity, YouTube history) and pause or delete it.
- **WhatsApp Privacy Settings:** Control who sees your last seen, profile photo, status, and read receipts under Settings → Privacy. Enable two-step verification for extra account security.
- **Facebook & Instagram:** Set your account to private, review "Who can see your posts," limit past post visibility, and check the "Apps and Websites" section to remove third-party app access you no longer use.
- **Browser Privacy:** Regularly clear cookies, use private/incognito browsing for sensitive searches, and review site permissions for camera/microphone/location access.

Part C — How Data Sharing Really Works

- **First-Party Sharing:** Data you give directly to a company (e.g., your name and address to an e-commerce site to deliver a package).
- **Third-Party Sharing:** The company shares your data with partners — advertisers, analytics firms, or affiliated businesses — often buried in long Terms & Conditions.

- **Data Brokers:** Companies whose entire business is collecting, combining, and selling personal data profiles to advertisers or other businesses, usually without your direct knowledge.
- **App Permissions as a Data Pipeline:** Every permission you grant (location, contacts, microphone) is a potential ongoing data-sharing channel, not a one-time action.
- **Cross-Border Data Transfer:** Under the DPDP Act, 2023, the Indian government can restrict transfer of personal data to specific countries, rather than allowing unrestricted global transfer.

Part D — Digital Footprints

A **digital footprint** is the trail of data you leave behind through your online activity.

Type	Description	Examples
Active Footprint	Data you deliberately share	Posts, comments, photos, reviews, forum replies
Passive Footprint	Data collected without your direct action	Cookies, IP address logs, location tracking, ad-tracking pixels

Your digital footprint is often **permanent**. Deleted posts can be archived, screenshotted, or cached by search engines before removal. Employers, colleges, and even loan/visa officers increasingly review applicants' digital footprints before making decisions.

Important Terminologies

Term	Simple Meaning
PII (Personally Identifiable Information)	Any information that can identify a specific individual
Cookie	A small file websites store on your device to remember your activity
Metadata	"Data about data" — e.g., the time, location, and device used to take a photo
Data Breach	Unauthorised access to or leak of stored personal data
Encryption	Converting data into unreadable code so only authorised parties can read it
Anonymisation	Removing identifying details so data can no longer be traced to a person
Consent Manager	A registered platform to manage what data-sharing consents you've given
Data Fiduciary	The organisation deciding why/how your data is processed
Data Principal	You — the person the data belongs to
Masked Aadhaar	An Aadhaar copy showing only the last 4 digits of your number
Virtual ID (VID)	A temporary, revocable 16-digit number used instead of your Aadhaar number
Data Minimisation	Collecting only the data that is truly necessary

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Step-by-Step Understanding

How Your Data Travels From You to an Advertiser

1. **Collection** — You install an app and grant permissions (location, contacts)
2. **Processing** — The app's servers collect and analyse your behaviour
3. **Sharing** — Data is shared with advertising or analytics partners under the Terms of Service
4. **Profiling** — Your interests, habits, and demographics are combined into an advertising profile
5. **Targeting** — Ads are served to you (and sold to advertisers) based on this profile

How to Run a Personal Privacy Audit (Step-by-Step)

1. Open your phone's app permissions list and revoke anything unnecessary
2. Run Google's/Apple's privacy/security checkup tool
3. Review privacy settings on each social media account
4. Search your own name online to see what is publicly visible
5. Check your Aadhaar authentication history on the UIDAI portal
6. Delete old, unused accounts and apps you no longer use
7. Update passwords and enable Two-Factor Authentication everywhere

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Real-Life Indian Examples

(Widely reported public matters, described for educational awareness.)

- **Justice K.S. Puttaswamy vs Union of India (2017):** A retired judge challenged the mandatory nature of Aadhaar, leading a nine-judge Supreme Court bench to unanimously declare privacy a fundamental right under Article 21 — one of the most significant judgments in India's digital rights history.
- **WhatsApp's 2021 Privacy Policy Update:** A proposed update on data sharing with its parent company sparked massive public debate in India over consent and data sharing, leading many users to explore alternative apps and prompting closer regulatory attention to messaging-app privacy practices.
- **Cambridge Analytica Case (Global, with Indian Relevance):** Millions of Facebook users' data, including Indian users, was harvested without proper consent and reportedly used for political profiling — a global wake-up call about third-party data sharing.

Practical Scenarios

Scenario 1: A free "flashlight" app on your phone asks for permission to access your contacts and location.
A flashlight has no legitimate need for your contacts or location. Deny the permission or avoid the app.

Scenario 2: You receive a "KYC update" message asking you to share your Aadhaar and bank details to keep your SIM active.
Telecom and bank KYC updates are never done by sharing details over SMS/WhatsApp links. Verify only via the official app or by visiting a branch/store.

Scenario 3: Your child wants to create a social media account and the app asks for a birth date.
Under the DPDP Act, anyone under 18 is a "child," and platforms must obtain verifiable parental consent before processing their data. Get involved in reviewing the account's settings before allowing sign-up.

Scenario 4: You want to sell your old smartphone.
Perform a full factory reset and remove your Google/Apple account first — a factory reset alone does not always guarantee that recoverable personal data is completely gone if you skip this step.

Government Rules / Guidelines

- **Data Protection Board of India (DPBI):** The independent body established under the DPDP Act, 2023, to monitor compliance, direct breach remediation, and hear grievances from citizens.
- **MeitY (Ministry of Electronics and Information Technology):** The nodal ministry responsible for notifying the DPDP Rules and overseeing India's overall data governance framework.
- **UIDAI (Aadhaar):** Provides tools such as **masked Aadhaar** (hides the first 8 digits), **biometric lock/unlock**, a **Virtual ID (VID)** as an alternative to sharing your Aadhaar number, and an **Aadhaar Authentication History** log you can check for unauthorised use.
- **CERT-In:** Issues advisories on data-handling security practices and coordinates responses to major data breaches affecting Indian users.

Laws & Policies

Law / Provision	What It Covers
Article 21, Constitution of India	Right to life and personal liberty — interpreted by the Supreme Court in *Puttaswamy (2017)* to include the fundamental right to privacy
Digital Personal Data Protection Act, 2023	India's comprehensive personal data law — rights of Data Principals, obligations of Data Fiduciaries, Consent Managers, and the Data Protection Board
DPDP Rules, 2025	Notified 13 November 2025; phased rollout — Consent Manager registration from November 2026, most compliance obligations enforceable from May 2027
DPDP Act — Children's Data (Section 9)	Anyone under 18 is a "child"; platforms must obtain verifiable parental consent and are barred from tracking, behavioural monitoring, and targeted advertising directed at children (with narrow, notified exemptions)
IT Act, 2000 — Section 43A	Compensation liability for a body corporate that fails to implement reasonable security practices to protect sensitive personal data
IT Act, 2000 — Section 72	Penalty for breach of confidentiality and privacy by a person who accesses data under lawful authority and discloses it without consent
IT Act, 2000 — Section 72A	Punishment for disclosure of personal information in breach of a lawful contract
Aadhaar Act, 2016	Governs the collection, storage, and use of Aadhaar data, restricting sharing without consent

Data Fiduciaries that process personal data of **10 million or more** Data Principals, or handle large-scale sensitive data, may be notified as **Significant Data Fiduciaries (SDFs)** — facing extra obligations like appointing a Data Protection Officer and conducting regular audits.

Important Statistics

Metric	Figure
Internet users in India (end of 2025)	Approximately 1.03 billion (about 70% penetration)
Social media user identities in India (Oct 2025)	Approximately 500 million (about 34% of population)
Parents aware of online privacy risks (2025 survey)	Around 60%
Parents who regularly adjust privacy settings	Around 43%
Parents who never use privacy settings at all	Nearly one-third
Puttaswamy judgment bench size	9 judges (unanimous verdict, August 24, 2017)
SDF threshold (illustrative criterion)	10 million+ Data Principals processed
DPDP Rules notification date	13 November 2025

(Figures are approximate and drawn from published government notifications, court records, and industry survey reports as of 2025–2026.)

Best Practices

- Review app permissions regularly and revoke anything unnecessary
- Use strong, unique passwords with a password manager
- Enable Two-Factor Authentication on email, banking, and social accounts
- Use masked Aadhaar or Virtual ID instead of sharing your full Aadhaar number
- Read privacy policies at least briefly before accepting, focusing on data-sharing clauses
- Regularly check your Aadhaar Authentication History for unfamiliar activity
- Log out of and delete unused old accounts and apps
- Set social media accounts to private by default

Common Mistakes

- Accepting all app permissions without reviewing them
- Posting sensitive documents (Aadhaar, PAN, boarding passes, tickets) on social media

- Reusing the same password across multiple platforms
- Ignoring privacy policy updates from apps you use daily
- Sharing OTPs or personal documents for "KYC updates" via unofficial links
- Leaving old, unused accounts active indefinitely
- Assuming "deleted" content is permanently gone

Myths vs Facts

Myth	Fact
"I have nothing to hide, so privacy doesn't matter to me"	Privacy protects everyone from misuse, discrimination, and fraud — not just those with "something to hide"
"Deleting a post removes it from the internet completely"	Cached copies, screenshots, and archives can preserve content even after deletion
"Privacy policies are just legal formalities I can skip"	Privacy policies often reveal exactly how and with whom your data will be shared
"Sharing my Aadhaar number is always safe since it's just an ID"	Aadhaar numbers combined with other leaked data can enable identity theft; use masked Aadhaar or VID where possible
"Free apps don't cost me anything"	Free apps often monetise by collecting and sharing user data — you pay with your privacy
"Only large companies need to worry about data protection law"	Under the DPDP Act, any entity processing personal data — including small businesses and individuals in certain contexts — has obligations

Do's and Don'ts

Do	Don't
Review and limit app permissions regularly	Don't grant every permission an app requests by default
Use masked Aadhaar/Virtual ID when sharing ID proof	Don't post your full Aadhaar or PAN details on social media
Read the data-sharing section of privacy policies	Don't blindly click "Accept All" without any review
Set social profiles to private	Don't leave profiles fully public without considering who can see them
Verify identity requests independently	Don't share OTPs or documents through unsolicited links

Safety Tips

For Social Media
- Set profiles to private and review follower/connection requests carefully
- Turn off precise location tagging on posts
- Regularly review which third-party apps have access to your account

For Aadhaar & KYC
- Use masked Aadhaar or Virtual ID (VID) wherever full Aadhaar isn't strictly required
- Lock your Aadhaar biometrics when not actively using them
- Periodically check your Aadhaar Authentication History on the UIDAI portal

For Families & Children
- Review and set up parental consent and privacy controls before a child creates any online account
- Talk openly with children about what is safe to share online
- Avoid posting identifiable details of children (school name, location) publicly

For Everyday Apps & Devices
- Turn off microphone/camera/location access for apps that don't need it
- Regularly clear cookies and review browser site permissions
- Factory reset and remove your account before selling or discarding a device

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Practical Checklist

- ☐ Reviewed and limited app permissions on my phone
- ☐ Ran a Google/Apple privacy checkup
- ☐ Set social media accounts to private
- ☐ Used masked Aadhaar/VID instead of full Aadhaar where possible
- ☐ Checked Aadhaar Authentication History for unfamiliar activity
- ☐ Enabled Two-Factor Authentication on key accounts
- ☐ Removed unused old apps and accounts
- ☐ Talked to my family about safe data-sharing habits

Quick Revision Notes

- Digital identity = the sum of all data points that represent you online
- Privacy = your right to control who accesses that identity
- Core principles: Notice, Consent, Purpose Limitation, Data Minimisation, Security, Accountability
- Data Principal (you), Data Fiduciary (organisation), Consent Manager, Data Protection Board — the four key roles under India's DPDP Act

- Digital footprints are active (what you post) or passive (what's tracked) — and often permanent
- Puttaswamy (2017) made privacy a fundamental right under Article 21
- DPDP Act 2023 + Rules 2025 govern personal data protection in India, phased rollout through May 2027

Key Takeaways

1. Your personal data has real value — treat it as carefully as you treat your money
2. Privacy is a fundamental right in India, not a courtesy extended by companies
3. Every app permission is an ongoing data-sharing decision, not a one-time click
4. Digital footprints can last far longer than you expect — think before you post
5. Simple habits — reviewing permissions, using masked Aadhaar, setting profiles private — prevent most privacy harms

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Glossary

- **Anonymisation** — Removing identifying details so data can't be traced to a person
- **Consent Manager** — A registered platform to manage your data-sharing consents
- **Cookie** — A small file that lets websites remember your activity
- **Data Breach** — Unauthorised access to or leak of stored personal data
- **Data Fiduciary** — The entity deciding why/how personal data is processed
- **Data Principal** — The individual to whom personal data belongs
- **Data Protection Board of India** — India's independent data-protection regulator
- **Digital Footprint** — The trail of data left behind by online activity
- **Encryption** — Converting data into unreadable code for protection
- **Masked Aadhaar** — An Aadhaar copy showing only the last 4 digits
- **Metadata** — Data describing other data (e.g., time/location of a photo)
- **PII** — Personally Identifiable Information
- **Significant Data Fiduciary (SDF)** — A fiduciary facing extra obligations due to scale/sensitivity of data processed
- **Virtual ID (VID)** — A temporary, revocable number used instead of your Aadhaar number

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Frequently Asked Questions

1. What is digital identity?
The complete online representation of you, built from data like your government IDs, social profiles, and browsing behaviour.

2. Is privacy a legal right in India?
Yes. The Supreme Court's 2017 Puttaswamy judgment declared privacy a fundamental right under Article 21.

3. What is the DPDP Act, 2023?
India's comprehensive law governing how personal data is collected, processed, and protected, with rights for individuals and obligations for organisations.

4. What is a Data Principal?
The individual whose personal data is being processed — essentially, you.

5. What is a Data Fiduciary?
The organisation or entity that decides the purpose and means of processing your personal data.

6. What is masked Aadhaar?
An Aadhaar copy that shows only the last 4 digits, hiding the first 8 for extra privacy.

7. What is a Virtual ID (VID)?
A temporary, revocable 16-digit number you can use instead of sharing your actual Aadhaar number.

8. Can I ask a company to delete my data?
Yes, under the DPDP Act, Data Principals have the right to seek correction and erasure of their personal data.

9. What is a digital footprint?
The trail of data — active (posts) or passive (tracking) — left behind by your online activity.

10. Can deleted social media posts really disappear completely?
Not always — cached copies, screenshots, and archives can preserve content even after deletion.

11. Do children need special protection under India's data law?
Yes. Anyone under 18 is legally a "child," and platforms must get verifiable parental consent before processing their data.

12. What is a Consent Manager?
A registered, independent platform through which you can give, review, or withdraw consent for data sharing.

13. What happens if a company suffers a data breach?
It must notify affected individuals and the Data Protection Board of India under the DPDP Act.

14. Are all app permissions necessary?
No. Many apps request more permissions than they actually need to function — always review before granting.

15. What is the difference between anonymisation and encryption?
Anonymisation removes identity from data; encryption scrambles data so only authorised parties can read it.

16. Is it safe to post my Aadhaar card on social media?
No. UIDAI advises against posting Aadhaar details publicly; use masked Aadhaar when sharing is necessary.

17. What is a Significant Data Fiduciary?
An organisation processing data at a large scale or of high sensitivity, facing extra compliance obligations like audits and a Data Protection Officer.

18. Can I check who has used my Aadhaar for authentication?
Yes, via the Aadhaar Authentication History feature on the UIDAI portal.

19. What should I do before selling my old phone?
Back up your data, sign out of all accounts, and perform a full factory reset.

20. Who enforces India's data protection law?
The Data Protection Board of India, with appeals going to TDSAT (Telecom Disputes Settlement and Appellate Tribunal).

21. Are free apps really "free"?
Often not in the way you'd expect — many monetise by collecting and sharing your data with advertisers.

22. Where can I complain about misuse of my personal data?
You can approach the Data Fiduciary first, and if unresolved, file a complaint with the Data Protection Board of India.

Suggested Infographics, Flowcharts & Illustrations

(For JanGyan design team — placement suggestions only; no images generated here)

- **"Categories of Personal Data" icon wheel** → within "Complete Detailed Explanation, Part A"
- **Flowchart: "How Your Data Travels to an Advertiser"** → after "Step-by-Step Understanding"
- **Side-by-side comparison: Active vs Passive Digital Footprint** → within "Digital Footprints" section
- **Roles diagram: Data Principal → Data Fiduciary → Consent Manager → Data Protection Board** → after "Fundamental Concepts"
- **Masked Aadhaar vs Full Aadhaar visual comparison** → within "Government Rules/Guidelines"
- **Privacy Settings checklist visual card** → near "Practical Checklist"

Official References

- Digital Personal Data Protection Act, 2023, and Digital Personal Data Protection Rules, 2025 — Ministry of Electronics and Information Technology (MeitY)
- Supreme Court of India — Justice K.S. Puttaswamy (Retd.) vs Union of India (2017) judgment records
- Unique Identification Authority of India (UIDAI) — Aadhaar Usage Do's and Don'ts, Biometric Lock/Unlock and Virtual ID guidance
- Information Technology Act, 2000 (as amended), India Code portal
- Press Information Bureau (PIB), Government of India — releases on the DPDP Act and Rules
- Data Protection Board of India — official functions and procedures

Official Government Website Links

- MeitY (Data Protection): **https://www.meity.gov.in**
- UIDAI (Aadhaar Services): **https://uidai.gov.in**
- myAadhaar Portal: **https://myaadhaar.uidai.gov.in**
- CERT-In: **https://www.cert-in.org.in**
- Press Information Bureau: **https://pib.gov.in**
- National Cyber Crime Reporting Portal: **https://cybercrime.gov.in**
- Digital India: **https://www.digitalindia.gov.in**

Cybercrime Helpline (24×7, Toll-Free): 1930

Conclusion

Your digital identity is one of the most valuable things you own — and unlike your phone or your wallet, it can't simply be replaced if lost or misused. Understanding personal data, mastering your privacy settings, knowing how your data gets shared, and being mindful of your digital footprint puts you back in control.

Privacy is not about having something to hide — it is about having the right to choose. This is only the next step in your Cyber Safety journey with JanGyan. Stay alert, stay informed, and remember: **your data, your rules.**

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